



Habib University

EE-172/CS-130/CE-222 Digital Logic and Design – Spring 2026

Time = 30 minutes

Quiz 01 CLO 1
(Cog-3)

Max Points: 20

Question 1 [4 points]: Convert the following:

a) $(D5)_{20}$ to Base 10 [1 point]

$$(13 \cdot 20^1) + (5 \cdot 20^0) \\ = 265$$

b) $(11.3)_4$ to Base 10 [1 point]

$$(1 \cdot 4^1) + (1 \cdot 4^0) + (3 \cdot 4^{-1}) \\ = 5.75$$

c) $(11.3)_4$ to Base 2 [2 point]

$$11.3 \text{ base } 4 = 5.75 \text{ base } 10$$

$$(5)_{10} = (101)_2$$

$$0.75 \cdot 2 = 1.5$$

$$0.5 \cdot 2 = 1.0$$

$$(5.75)_{10} = (101.11)_2$$

Question 2 [4 points]: Perform the following operation in **BCD (Binary-Coded Decimal)** system and verify:

$$(2.8)_{10} + (6.51)_{10}$$

$$2.80 : 0010' . 1000' 0000$$

$$6.51 : 0110 . 0101 0001$$

$$1001 . 1101' 0001$$

$$+0110$$

$$= 1 0011$$

$$9.31 : 1001 . 0011 0001$$

Question 3 [4 points]: Convert the following numbers:

a) $(-48)_{10}$ to its signed magnitude representation (8bits) [2 points]

$$48 = (0110000)_2, 7 \text{ bits}$$

$$-48 = (10110000)_2, 8 \text{ bits}$$

b) $(-28)_{10}$ to its 2's complement representation (8bits) [2 points]

$$(28)_{10} = (00011100)_2$$

$$1's \text{ comp} = 11100011$$

$$2's \text{ comp} = 11100100$$

Question 4 [4 points]: Perform the following operations in the number system given:

a) $(33)_8 + (77)_8$ [2 points]

$$\begin{array}{r} 33 \\ + 77 \\ \hline 132 \end{array}$$

b) $(DE)_{16} - (89)_{16}$ [2 points]

$$\begin{array}{r} DE \\ - 89 \\ \hline 55 \end{array}$$

Question 5 [4 points]: An 8-bit register stores 10010101. Determine what value is stored in decimal in the register if the bits represent:

- a) Unsigned number
- b) Signed number (2's complement system)
- c) BCD

a) $(149)_{10}$

b) $(-107)_{10}$

c) $(9\ 5)_{10}$